

Nestor I. Ospina Gaitan

+57(300)292-4631

Bogota, Colombia

nestorivan.o@hotmail.com

Robotics and AI Software Engineer

GitHub: niospinag

LinkedIn: nestor-ospina

PROFESSIONAL SUMMARY

Robotics and AI Software Engineer with an M.Sc. in Industrial Automation and a strong foundation in Machine Learning, ROS2, and data engineering. Proven ability to design AI predictive models, orchestrate large-scale data pipelines (~1 TB), and develop decentralized control systems for heterogeneous robot swarms. Adept at leveraging Python, GCP, and simulation tools (Gazebo) to bridge the gap between AI algorithms and real-world autonomous navigation systems.

TECHNICAL SKILLS

Robotics & Vision	ROS/ROS2 (Nav2, Humble/Foxy), OpenCV, ArUco Markers, Gazebo, Coppeliasim, Swarm Robotics.
Programming Languages	Python (Advanced), SQL (Advanced), Bash/Shell, C++ (Familiar).
AI, Data & Cloud	Machine Learning (Random Forest), GCP (BigQuery), SQL Server, Looker Studio.
Hardware & Embedded	NVIDIA Jetson, Raspberry Pi, PCB troubleshooting, 3D Printing (Cura, SolidWorks).
Tools & DevOps	Linux (Ubuntu), Git, Jira, Advanced Excel.

WORK EXPERIENCE

DATA ANALYST / MACHINE LEARNING

May 2026 — Present

Banco Falabella

Bogotá, Colombia

- Engineered and deployed Random Forest machine learning models in Python to optimize business goal distribution, significantly increasing branch commission success rates from 50% to 85%.
- Architected scalable data pipelines using GCP (BigQuery) and SQL Server to process large-scale historical datasets (~1 TB), developing complex Stored Procedures (SPs) and use cases.
- Designed automated, real-time dashboards in Looker Studio and utilized advanced Excel (ODBC integrations) to translate complex data into actionable intelligence for C-level executives.

EXTERNAL ROBOTICS ADVISOR

Apr 2026 — Present

Universidad Santo Tomás

Bogotá, Colombia

- Act as a consultant and mentor for engineering students, providing expert guidance on robotics, embedded systems, and hardware troubleshooting.

ROBOTICS SOFTWARE ENGINEER

May 2024 — Aug 2025

Unlimited Robotics

Bogotá, Colombia

- Supported the maintenance and integration of the ROS2 Nav2 stack for differential-drive robots, ensuring platform stability for autonomous navigation tasks.
- Developed and tested ROS2 nodes in Python and C++ to process sensor data and orchestrate robot behaviors, validating control logic through high-fidelity Gazebo simulations.
- Managed Linux-based environments (Ubuntu), utilizing Bash scripting to automate background processes, deployment pipelines, and reliable startup sequences.

RELIABILITY ENGINEER (Field Operations)

Dec 2023 — Apr 2024

Kiwicampus S.A.S. (Kiwibot)

Orono, Maine, USA

- Slashed hardware iteration time by establishing a 24-hour rapid prototyping pipeline (SolidWorks, Cura, 3D printing) to design, print, and deploy custom modifications directly onto the autonomous delivery robot fleet.
- Diagnosed and resolved critical hardware/software failures in harsh weather conditions, troubleshooting complex PCB and sensor issues to maximize fleet uptime.

CAREER SABBATICAL (English Immersion)

Jul 2021 — Nov 2023

Cultural Exchange Program

USA

- Participated in US cultural exchange programs, achieving professional working proficiency in English and cross-cultural communication skills.

GRADUATE RESEARCH ENGINEER

Jan 2020 — Jun 2021

Bethune-Cookman University & UNAL

Daytona Beach, FL, USA

- Architected a decentralized multi-agent testbed consisting of 23 heterogeneous robots (15 small, 4 medium, 2 large rovers, and 2 drones) to evaluate swarm behavior and Game Theory algorithms.
- Developed a precise vision/localization pipeline using Python, OpenCV, and ArUco markers, integrating software with Raspberry Pi hardware for real-world validation.
- Co-authored a peer-reviewed scientific paper detailing the findings of the multi-agent control strategies (available on Google Scholar).

EDUCATION

Master's Degree in Industrial Automation | Universidad Nacional de Colombia

Jun 2021

Bachelor's Degree in Electrical Engineering | Universidad Nacional de Colombia

Jun 2018